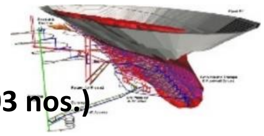
Indian Institute of Technology (Indian School of Mines) Dhanbad
Department of Mining Engineering
Midsem Examination Monsoon Semester 2024-2025
Course for 3rd Year (5th Semester) B.Tech. (Mining Engineering) Students (103 nos.)
Subject: Mine Planning and Economics (Code: MNC 301)



Time: 180 minutes (09:00 AM–12:00 PM) **|** **Venue:** NLHC LH 09-11 **|** **Date:** 28-11-2024 **|** **Maximum Marks:** 96

Answer all questions of the three sections and marks are assigned against each section
***Focus** on solving questions you **know first**, stay calm, **manage time wisely** and put forth the best effort*

Section A: Numerical Answer Type Questions – 60 marks

*Answer all the questions and **mark**s are assigned against each question*

1. **(a)** A machine costing Rs. 20,000/- was purchased and installed. Installation charges are Rs. 5,000/-. If scrap value is Rs. 5,000/- and the useful life is about 10 years, calculate the yearly depreciation by straight line method.

Total cost is Rs. 20,000/- + Rs. 5,000/- = Rs. 25,000/-

Depreciation charges = $\frac{25000-5000}{10} = 2000\ per\ year$

(b) A colliery has a total of 32 million metric tons in-situ reserves of coal. The recovery during mining is only 60%. If the production of coal from the mine is planned at 1.0 lakhs metric tons per month, estimate the life of the mine.

$$0.6 * x = 100000$$

$$x = \frac{1000000}{6}$$

$$Life\ of\ the\ mine = \frac{32000000}{\frac{1000000}{6}} = 32 * 6 = 192\ months = 16\ years$$

(c) Following sampling data from a drive is furnished:
Sample 1: 25 cm @ 3.2% Cu; **Sample 2:** 45 cm @ 2.2% Cu; **Sample 3:** 10 cm @ 3.1% Cu;
Cut-off grade is 2.0%. Minimum stoping width is 1 m. Calculate the average grade of the sample.

Since the cut-off grade is 2.0%, therefore, no sample can be rejected.

$$Average\ grade = \frac{25 * 3.2 + 45 * 2.2 + 10 * 3.1}{25 + 45 + 10} = 2.625$$

But, this grade is for 80 cm. Therefore, for 1 m = 100 cm width, the % grade will be less as 20 m barren rock is now mined for mining stoping width.

For 100 cm width = $\frac{2.625*80}{100} = 2.1\%$

Weighted average grade = 2.1%

(d) When the nominal rate of interest is 12 % and frequency of compounding is six times a year, estimate the difference between effective rate of interest and nominal rate of interest.

$$CI = P\left(1 + \frac{0.12}{6}\right)^{1*6} - P = 1.12616\ P - P = 0.12616P$$

Difference = 0.616%

(e) A mine is expected to earn Rs. 8,000/- per year for 10 years. A certain investor provided interest @ 9% per annum on the earning. What is the present value and sum of payment?

Sum of Payment = Earnings per year × Number of years = 8,000 × 10 = 80,000

$$\boldsymbol{PV = FW\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] = 8000\left[\frac{(1+0.09)^{10} - 1}{0.09(1+0.09)^{10}}\right] = 51341}$$

(f) Find the revenue per week that will maximize the company savings: $\boldsymbol{P = 2000 - 5Q}$, where $\boldsymbol{P}$ is price per unit and $\boldsymbol{Q}$ is demand per week.

R=P·Q

$$R = 2000Q - 5Q^2$$
$$\frac{dR}{dQ} = 2000 - 10Q = 0$$
$$Q = 200$$
$$\frac{d^2R}{dQ^2} = -10 < 0$$

Since, $\frac{d^2R}{dQ^2} < 0$, R is maximized at Q=200

Substitute Q=200 into the revenue equation:

$$R = 2000Q - 5Q^2 = 2000 * 200 - 5 * 200^2 = 200000$$

The maximum revenue per week is **₹200,000**, achieved when the demand is **200 units per week**.

(g) Calculate the amount of stowing material/day if daily productions from a copper mine using cut-and-fill stoping is 1500 tonnes. The stowing material of 1.6 t/m^3 is to be used. Ore density is 3 t/m^3. Consider the value of K factor as 0.7. Use the following relation $\boldsymbol{P'} = \frac{\gamma'}{\gamma}\boldsymbol{KP}$; where $\boldsymbol{P'}$= weight of stowing material to be used, **P** = weight of mineral (ore) to be extracted in tons. $\boldsymbol{\gamma'}$ = density of coal/ore in tons/m^3, $\boldsymbol{\gamma}$ = density of stowing material in tons/m^3, **K** = factor of stowing.

$$\boldsymbol{P' = \frac{\gamma'}{\gamma}KP = \frac{3}{1.6} * 0.7 * 1500 = 1968.75\ tons}$$

(h) Mention the relationship among, r_r =real rate of return from a mining project (%), r_i = inflation rate over the entire life of the mine (%), and r_n =nominal rate of return (%).

$$\boldsymbol{(1 + r_r)(1 + r_i) = (1 + r_n)}$$

(i) For a shrinkage stope the following data values are given

In-situ tonnage : 9000 tonnes
In-situ grade : 5.2 g/tonne
Average grade of waste : 1.4 g/tonne
Loss of ore in the stope : 10%
Dilution : 20%

Estimate the grade at the mill-head in g/tonne?

Grade at the mill head = $\frac{(5.2*0.9*9000)+(1.4*0.2*9000)}{[(0.9*9000)+(0.2*9000)]} = \frac{44640}{9900}$ =4.51 g/tonne

OR

Loss of ore in the stope = 9000 * $\frac{90}{100}$= 900 tonnes

So, total ore recovered = 9000 - 900 = 8100 tonnes

Dilution in the ore = 9000 * $\frac{20}{100}$ = 1800 tonnes

So, total ore after dilution = 8100+ 1800 = 9900 tonnes

Insitu grade = 5.2 g/tonne

So, the total metal recovered from insitu tonnage = 8100 * 5.2 = 42120 g

Average grade of waste = 1.4 g/tonne

So, the total metal recovered from dilution = 1800 * 1.4 = 2520 g

Total metal recovered = 42120 + 2520=44640 g

Now, Grade at the mill head = $\frac{44640}{9900}$ =4.51 g/tonne

(2 + 2 + 2 + 2 + 2 + 2 + 2 + 1 + 3 = 18 marks)

2. There is a mineral reserve of 30.8 x 10^6 tons containing 7.8 x 10^6 tons of copper ore with an average grade of 0.92%. A market survey suggested that 5000 tons of copper metal can be sold every year. Calculate daily and annual milling and mining rate, mine life, total copper recovered and overall stripping ratio. Assume mill recovery = 70%, combined smelter/refinery recovery = 96% and operating days = 300 days/year.

(4 marks)

Let X% = cut-off grade = 0.92%

Y = Total ore tonnage = 7.8 x 10^6 tons
Mineral Reserve = Z = 30.8 x 10^6 tons [constitutes tonnage of ore Y and waste (Z-Y) together]

$$\textit{Milling rate} = \frac{5000\ tpy}{\frac{0.92}{100} * 0.70 * 0.96} = 808747\ tpy = 2696\ tpd$$

$$\textit{Mill/Mine life (years)} = \frac{\textit{Ore reserves (tons)}}{\textit{Ore production rate (tpy)}} = \frac{7800000}{808747} = 9.65\ years$$

$$\textit{Mining Rate} = \frac{\textit{Mineral reserve (tons)}}{\textit{Mine life (yrs)}} = \frac{30800000}{9.656} = 3191710\ tpy = 10640\ tpd$$

$$\textit{Overall SR} = \frac{Z - Y(\textit{overburden tonnage})}{Y\ (\textit{ore tonnage})} = \frac{30800000 - 7800000}{7800000} = 2.95$$

$$\textit{Copper recovered} = \mathbf{7.8 * 10^6 * 0.0092 * 0.70 * 0.96 = 48223\ tons = 5000 * 9.65}$$

3. Costs:

m = mining = $1/ton
c = concentrating = $2/ton
r = refining = $5/lb
f = fixed cost = $300/year

Selling Price:

s = $25/lb

Capacities: Here tons is used for short tons (st).

M = mining = 100 tons/year
C = concentrating = 50 tons/year
R = refining = 40 lbs/year

Quantities

Q_m = amount to be mined (tons)
Q_c = amount sent to the concentrator (tons)
Q_r = amount of concentrator product sent for refining (lbs)

Recovery (Yield): y = 1.0 (100%recovery is assumed).

Table: Initial mineral inventory

Grade (lbs/ton)	Quantity (tons)
0.0-0.1	100
0.1-0.2	100
0.2-0.3	100
0.3-0.4	100
0.4-0.5	100
0.5-0.6	100
0.6-0.7	100
0.7-0.8	100
0.8-0.9	100
0.9-1.0	100
Q_m	1000

(a) Consider a combination of mining rate of 100 tons/year and concentrator cutoff grade of 0.50 lbs/ton is used and 50 tons of material having an average grade of 0.75 lbs/ton sent to the concentrator every year. The other 50 tons would be sent to the waste dump. Since the concentrator capacity **C** is 50 tons/year, this is an acceptable situation. The concentrator product becomes the refinery feed. In this case it would be 37.5 lbs/year (0.75 lbs/ton × 50 tons/year). Since the refinery can handle 40 lbs/year, it would be operating at below its rated capacity R. Calculate the yearly profit and NPV assuming an interest rate of 15%.

This combination of mining rate and cutoff grade would yield a yearly profit $\boldsymbol{P_y}$ of

$$\boldsymbol{P_y = (25 - 5) * 37.5 - 2 * 50 - 1 * 100 - 300 = \$250}$$

These profits would continue for 10 years and hence the total profit would be $2500. The NPV assuming an interest rate of 15% would be

$$\boldsymbol{NPV = \frac{250 * [(1.15)^{10} - 1]}{0.15 * (1.15)^{10}} = \$1,\ 254.69}$$

(b) Prove the following relationships to estimate the cut-off grade for maximum profit, where symbols have their usual meaning.

(i) $g_m = \frac{c}{y(s-r)}$ by constraining mining rate

(ii) $g_c = \frac{c+\frac{f}{C}}{y(s-r)}$ by constraining concentration rate

(iii) $g_r = \frac{c}{y\left(s-r-\frac{f}{R}\right)}$ by constraining refining rate.

Using the definitions given in the preceding section, the basic equations can be developed. The total costs Tc are

$$T_c = mQ_m + cQ_c + rQ_r + fT \qquad (6.7)$$

Since the revenue R is

$$R = sQ_r \qquad (6.8)$$

the profit P is given by

$$P = R - T_c = sQ_r - (mQ_m + cQ_c + rQ_r + fT) \qquad (6.9)$$

Combining terms, yields

$$P = (s - r)Q_r - mQ_m - cQ_c - fT \qquad (6.10)$$

This is the basic profit expression.

It can be used to calculate the profit from the next Qm of material mined.

Calculate cutoff grade assuming that the mining rate is the governing constraint.

If the mining capacity M is the applicable constraint, then the time needed to mine material Q_m is

$$T_m = \frac{Q_m}{M} \qquad (6.11)$$

Equation (6.10) becomes

$$P = (s - r)Q_r - cQ_c - \left(m + \frac{f}{M}\right) Q_m \qquad (6.12)$$

To find the grade which maximizes the profit under this constraint one first takes the derivative of (6.12) with respect to g.

$$\frac{dP}{dg} = (s - r)\frac{dQ_r}{dg} - c\frac{dQ_c}{dg} - \left(m + \frac{f}{M}\right)\frac{dQ_m}{dg} \qquad (6.13)$$

Step 1. Determination of the economic cutoff grade – one operation constraining the total capacity

However, the quantity to be mined is independent of the grade, hence

$$\frac{dQ_m}{dg} = 0 \qquad (6.14)$$

Equation (6.13) becomes

$$\frac{dP}{dg} = (s - r)\frac{dQ_r}{dg} - c\frac{dQ_c}{dg} \qquad (6.15)$$

The quantity refined Qr is related to that sent by the mine for concentration Qc by

$$Q_r = \bar{g}yQ_c \qquad (6.16)$$

where $\bar{g}$ is the average grade sent for concentration, and y is the recovery.

Taking the derivative of Q_r with respect to grade one finds that

$$\frac{dQ_r}{dg} = \bar{g}y\frac{dQ_c}{dg} \qquad (6.17)$$

Substituting Equation (6.17) into (6.15) yields

$$\frac{dP}{dg} = [(s - r)\bar{g}y - c]\frac{dQ_c}{dg} \qquad (6.18)$$

The lowest acceptable value of $\bar{g}$ is that which makes

$$\frac{dP}{dg} = 0$$

Thus the cutoff grade gm based upon mining constraints is the value of g¯ which makes

$$(s - r)\bar{g}y - c = 0$$

Thus

$$g_m = \bar{g} = \frac{c}{(s-r)y} \qquad (6.19)$$

Substituting the values from the example yields

$g_m = \frac{c}{(s-r)y} = \frac{\$2}{(25-5)*1.0}$ = 0.10 lbs/ton

Calculate cutoff grade assuming that the concentrating rate is the governing constraint.

If the **concentrator capacity C** is the controlling factor in the system, then the time required to mine and process a Qc block of material (considering that mining continues simultaneously with processing) is

$$T_c = \frac{Q_c}{C} \qquad (6.20)$$

Substituting Equation (6.20) into (6.10) gives

$$P = (s - r)Q_r - mQ_m - cQ_c - f\frac{Q_c}{C} \qquad (6.21)$$

Rearranging terms one finds that

$$P = (s - r)Q_r - \left(c + \frac{f}{C}\right)Q_c - mQ_m$$

Differentiating with respect to g and setting the result equal to zero yields

$$\frac{dP}{dg} = (s - r)\frac{dQ_r}{dg} - \left(c + \frac{f}{C}\right)\frac{dQ_c}{dg} - m\frac{dQ_m}{dg}$$

As before $\frac{dQ_m}{dg} = 0$

$$\frac{dQ_r}{dg} = gy\frac{dQ_c}{dg}$$

Thus $(s - r)gy\frac{dQ_c}{dg} = \left(c + \frac{f}{C}\right)\frac{dQ_c}{dg}$

The cutoff grade when the concentrator is the constraint is

$$g_c = \frac{c + \frac{f}{C}}{y(s - r)} \qquad (6.22)$$

For the example, this becomes $g_c = \frac{2 + \frac{300}{50}}{1.0(25 - 5)} = 0.40$

Calculate cutoff grade assuming that the refining rate is the governing constraint.

$$T_r = \frac{Q_r}{R} \qquad (6.23)$$

Substituting (6.23) into Equation (6.10) yields

$$P = (s - r)Q_r - mQ_m - cQ_c - f\frac{Q_r}{R}$$

$$P = \left(s - r - \frac{f}{R}\right)Q_r - cQ_c - mQ_m \qquad (6.24)$$

Differentiating with respect to g and setting the result equal to zero gives

$$\frac{dP}{dg} = \left(s - r - \frac{f}{R}\right)\frac{dQ_r}{dg} - c\frac{dQ_c}{dg} - m\frac{dQ_m}{dg} = 0$$

Simplifying and rearranging gives

$$g_r = \frac{c}{y\left(s - r - \frac{f}{R}\right)} \qquad (6.25)$$

For this example $g_r = \frac{c}{y\left(s - r - \frac{f}{R}\right)}$ (6.26)

(c) After proving the above relationship for cut-off grade under different constraints, calculate the respective maximum profits and NPV assuming an interest rate of 15% using the above-mentioned details.

Mining rate limit: g_m = 0.10 lbs/ton; $\bar{g}$ = 0.55 lbs/tons

Q_r = 0.55 × 900 = 495 tons

$$T_m = \frac{1000}{100} = 10\ years$$

P_m = (\$25 – \$5)*495 – \$2 × 900 – \$1 × 1000 – \$300 × 10 = \$4100

$$NPV_m = \frac{P_m}{(1 + i)^n} = \frac{4100}{(1.15)^{10}} = 1013.48$$

Concentrator limit: g_c = 0.40 lbs/ton; $\bar{g}$ = 0.7 lbs/tons

Q_c = 0.60 × 1000 = 600 tons

C = 50 tons/year

T_c = 600 / 50 = 12 years

M= 1000 /12 = 83.3 tons/year

Q_r = 600 × 0.7 × 1.0 = 420 lbs

P_c = (25 – 5)420 – 2 × 600 – 1 × 1000 – 300 × 12 = \$2600

$$NPV_c = \frac{P_c}{(1 + i)^n} = \frac{2600}{(1.15)^{12}} = 485.95$$

Refinery limit: g_r = 0.16 lbs/ton; $\bar{g}$ = 0.58 lbs/tons

Q_c = 0.84 × 1000 = 840 tons

Q_r = 840 × 0.58 × 1.0 = 487.2 lbs

T_r = 487.2 / 40 = 12.18 years
M = 1000 12.18 = 82.1 tons/year
C = 840 12.18 = 69 tons/year
P_r = (25 − 5)487.2 − 2 × 840 − 1 × 1000 − 300 × 12.18 = $3410

$$NPV_c = \frac{P_c}{(1+i)^n} = \frac{3410}{(1.15)^{12.18}} = 621.52$$

(d) After balancing the operations, when operations are taken in combination, the optimum/economic cutoff grades are $g_{mc} = 0.50$, $g_{cr} = 0.60$, $g_{mr} = 0.456$. Determine the overall optimum of the six cutoff grades. Perform only one iteration process to calculate the profit per year and NPV assuming an interest rate of 15% using the above-mentioned details.

(2 + 6 + 4 + 4 = 16 marks)

The corresponding optimum grades for each pair (G_{mc}, G_{cr}, and G_{mr}) are selected using the following rules:

$$G_{mc} = \begin{cases} g_m & if\, g_{mc} \le g_m \\ g_c & if\, g_{mc} \ge g_c \\ g_{mc} & otherwise \end{cases}$$

$$G_{rc} = \begin{cases} g_r & if\, g_{rc} \le g_r \\ g_c & if\, g_{rc} \ge g_c \\ g_{rc} & otherwise \end{cases}$$

$$G_{mr} = \begin{cases} g_m & if\, g_{rc} \le g_n \\ g_r & if\, g_{rc} \ge g_r \\ g_{mr} & otherwise \end{cases}$$

The overall optimum cutoff grade G is just the middle value of G_{mc}, G_{mr}, and G_{rc}.
In our example the six possible cutoff grades are:

g_m = 0.10
g_c = 0.40
g_r = 0.16
$g_{mc} = \mathbf{0.50}$
g_{mr} = **0.456**

g_{cr} = **0.60**

Consider them in groups of three:

Group 1 g_m = 0.10, g_c = 0.40, $g_{mc} = \mathbf{0.50}$ — Choose the middle value G_{mc} = 0.40

Group 2 g_m = 0.10, g_r = 0.16, $g_{mr} = \mathbf{0.50}$ — Choose the middle value G_{mr} = 0.16

Group 3 g_r = 0.10, g_c = 0.40, $g_{cr} = \mathbf{0.60}$ — Choose the middle value G_{cr} = 0.40

Considering the three middle values
G_{mc} = 0.40
G_{mr} = 0.16
G_{cr} = 0.40
one chooses one numerically in the middle **G = 0.40 lbs/ton**
From Table 6.24, the average grade $\overline{g_c}$ of the material sent to the concentrator for a cutoff of 0.40 lbs/ton would be $\overline{g_c}$ = **0.70 lbs/ton**
For 100% recovery the quantities are
Q_m = 1000 tons
Q_c = 600 tons
Q_r = 420 lbs
Applying the respective capacities to these quantities one finds that

T_c = 600 / 50 = 12 years
T_r = 420 / 40 = 10.5 years
T_m = 1000 / 100 = 10 years
Since the concentrator requires the longest time, it controls the production capacity.
The total profit is P = \$2600 and the profit per year P_y is
P_y = \$2600 / 12 years = \$216.70/year
The net present value of these yearly profits assuming an interest rate of 15% is
$NPV = P_y \frac{(1+i)^{12}-1}{i(1+i)^{12}} = \frac{(1.15)^{12}-1}{0.15(1.15)^{12}}$ = \$1174.60
In summary:
the concentrator is the controlling production limiter;
concentrator feed = 50 tons/year;
optimum mining cutoff grade = 0.40 lbs/ton (constant);
total concentrator feed = 600 tons;
average concentrator feed grade = 0.70 lbs/ton;
years of production = 12 years;
copper production/year = 35 lbs;
total copper produced = 420 lbs;
total profits = \$2600.40;
net present value = \$1174.60.
The previous section considered the selection of the cutoff grade with the objective being to maximize profits.
In most mining operations today, the objective is to maximize the net present value NPV.
In this section the Lane approach to selecting cutoff grades maximizing NPV subject to mining, concentrating and refining constraints will be discussed.
For the example of the previous section a fixed mining cutoff grade of 0.40 lbs/ton was used.
One found that
Q_m = 83.3 tons/year
Q_c = 50 tons/year
Q_r = 35 lbs/year
f = \$300/year
Lifetime = 12 years
The yearly profit would be
P_j = 35.0 × \$20 − 50 × \$2 − 83.3 × \$1 − \$300 = \$216.70
when finding the maximum total profit, the profits realized in the various years are simply summed.
The total profit is, therefore
$P_T = \sum_{j=1}^{12} P_j = 12 * 216.7 =$ \$2600.40
The net present value for this uniform series of profits (Chapter 2) is calculated using

$$NPV = P_j \frac{(1+i)^n - 1}{i(1+i)^n}$$

$NPV = 216.7 * \frac{(1.15)^{12}-1}{0.15*(1.15)^{12}}$= \$1174.6

$$g_m = 0.10$$

$$g_c = \frac{c+\frac{f+Vd}{C}}{(s-r)y} = \frac{2+\frac{300+0.15*1174.6}{50}}{(25-5)*1.0} = 0.576$$

$$g_r = \frac{c}{\left(s - r - \frac{f+Vd}{R}\right)} = \frac{2}{\left(20 - \frac{300 + 0.15 * 1174.6}{40}\right)} = 0.247$$

g_m = 0.10; g_c = 0.576; g_r = 0.247
$g_{mc} = 0.50$; g_{rc} = 0.60; $g_{mr} =$ 0.456
Applying the rules to select the overall pair optimum yields

$\boldsymbol{G_{mc}}$ = 0.40; $\boldsymbol{G_{mr}}$ = 0.40; $\boldsymbol{G_{rc}}$ = 0.16

The overall optimum is the middle value **G = 0.50**

Returning to the grade distribution Table 6.24 one finds that the average grade is 0.75.

If the mining rate $\boldsymbol{Q_m}$ = 100, then $\boldsymbol{Q_c}$ = 50 and $\boldsymbol{Q_r}$ = 37.5.

Both the mine and the concentrator are at their rated capacities.

The profit in a given year is

$\boldsymbol{P = (s - r)Q_r - mQ_m - cQ_c - fT}$ **= 20 × 37.5 – 2 × 50 – 1 × 100 – 300 = $250**

The number of years is

$\boldsymbol{n = \frac{Q}{Q_m}}$ = $\boldsymbol{\frac{1000}{100}}$ = 10 years

The present value becomes

$\boldsymbol{V = \frac{250*[(1.15)^{10}-1]}{0.15*(1.15)^{10}}}$ = $1254.7

This becomes the third estimate for $\boldsymbol{V}$ to be used in calculating $\boldsymbol{g_c}$ and $\boldsymbol{g_r}$.

$$\boldsymbol{g_c = \frac{2 + \frac{300 + 0.15 * 1254.7}{50}}{(25 - 5) * 1.0} = 0.588}$$

$$\boldsymbol{g_r = \frac{2}{\left(20 - \frac{300 + 0.15 * 1254.7}{40}\right)} = 0.247}$$

The six possible values are: $\boldsymbol{g_m}$ **= 0.10;** $\boldsymbol{g_c}$ **= 0.588;** $\boldsymbol{g_r}$ **= 0.257;** $\boldsymbol{g_{mc} = 0.50;}$ $\boldsymbol{g_{rc}}$ **= 0.60;** $\boldsymbol{g_{mr} =}$ **0.456**

The optimum pairs are: $\boldsymbol{G_{mc}}$ **= 0.50;** $\boldsymbol{G_{mr}}$ **= 0.257;** $\boldsymbol{G_{rc}}$ **= 0.588**

and the overall optimum (G = 0.50) is the same as found with the previous estimate. Hence in year 1

Optimum cutoff grade = 0.50 lbs/ton

Quantity mined = 100 tons

Quantity concentrated = 50 tons

Quantity refined = 37.5 lbs

Profit = $250

4. A grade block model of molybdenum sulphide (MoS_2) has been prepared using blocks 50 ft × 50 ft × 50 ft. The tonnage factor is 12.5 ft^3/st, hence, each block contains 10,000 tons. The following data were used in preparing the grade block model and in running the floating cone. The atomic weight of molybdenum (Mo) and sulfur (S) is 95.96 amu and 32.06 amu respectively.

Mining cost = $0.74/st

Processing cost = $1.89/st ore

General and administrative (G&A) cost = $0.67/st ore

Mill recovery = 90%

Selling price = $6/lb contained molybdenum (F.O.B. mill site)

Calculate the breakeven grade (%) for MoS_2 and also, compare the breakeven analysis for a price of $2/lb and $1.5/lb molybdenum.

(4 marks)

Since one pound of MoS_2 contains 0.60 lbs Mo, the equivalent price is $3.24/lb MoS_2.

In determining the final pit limits, the costs and revenues involved in mining and processing 1 ton of material containing X% MoS_2 are first determined.

Cost ($/st) = 0.74 + 0.67 + 1.89 = $3.30/st

Revenue ($/st) = $\frac{X}{100} * 2000 * 0.90 * \$6.00 * 0.60 = 64.8X$

Equating costs and revenues

64.8X = 3.30 one

One finds that the breakeven grade (X%) is

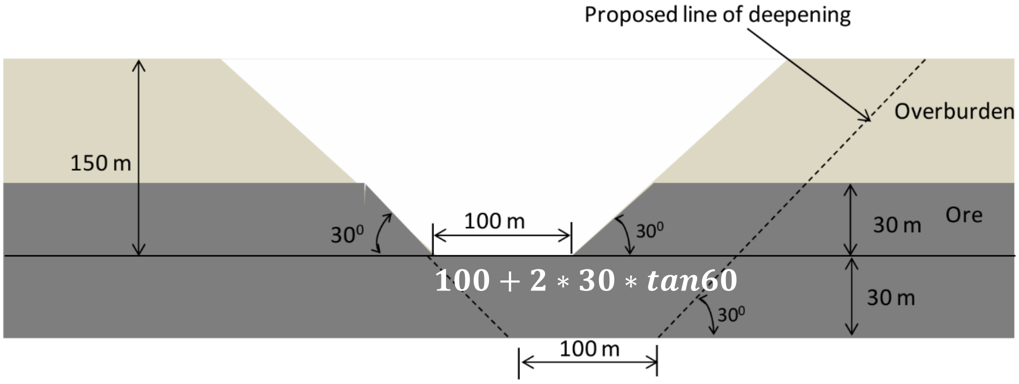
X = 0.05% MoS_2

This cutoff grade corresponds to a commodity price of $6/lb contained molybdenum.

If a price of $2/lb Mo is selected instead, then by redoing the breakeven analysis one finds that X = 0.15%

For a price of $1.50/lb Mo, the breakeven grade is X = 0.20%

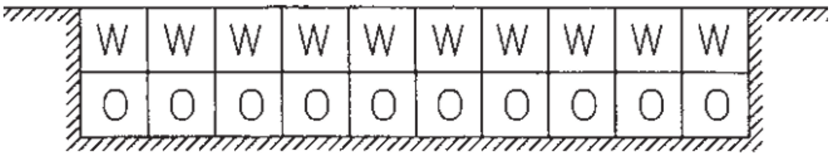
5. An operating surface mine is proposed to be deepened by 30 m as shown in the figure. If the density of the ore is 2.4 tonne/m^3, estimate the incremental stripping ratio (in m^3/tonne) for the deepening.



$$\frac{120 * 60tan60 = OB\ volume}{2.4\left[\left(\frac{1}{2} * 30 * (200 + 60tan60)\right) + (30 * 60tan60)\right] = ore\ tonnage} = 0.676\ m^3/t$$

(3 marks)

6. Consider the property shown in the figure below which has 10 ore blocks overlain by 10 waste blocks. A production rate of 4 blocks per year (irrespective of whether the blocks are ore or waste) will be assumed. The net value for an ore block is Rs. 2 and the cost of removing the waste is Rs. 1/block. The total cost involved in waste removal would be Rs. 10 and the ore value is Rs. 20. If both ore and waste could be mined instantaneously, the net present value would be Rs. 10. However, it is not possible to extract all of these 20 blocks at once due to practical constraints. Discuss any four combinations of scheduling scenarios which may be considered. Also, mention the most favorable combination out of them based on NPV (@10%).



(4 marks)

(1) 4W→4W→2W2O→4O→4O

$$NPV = -\frac{4}{(1.1)^1} - \frac{4}{(1.1)^2} + \frac{2}{(1.1)^3} + \frac{8}{(1.1)^4} + \frac{8}{(1.1)^5} = Rs. 4.99$$

(2) 3W1O→3W1O →3W1O →1W3O→4O

$$NPV = -\frac{1}{(1.1)^1} - \frac{1}{(1.1)^2} - \frac{1}{(1.1)^3} + \frac{5}{(1.1)^4} + \frac{8}{(1.1)^5} = Rs. 5.89$$

(3) 2.5W1.5O→2.5W1.5O →2.5W1.5O →2.5W1.5O→4O

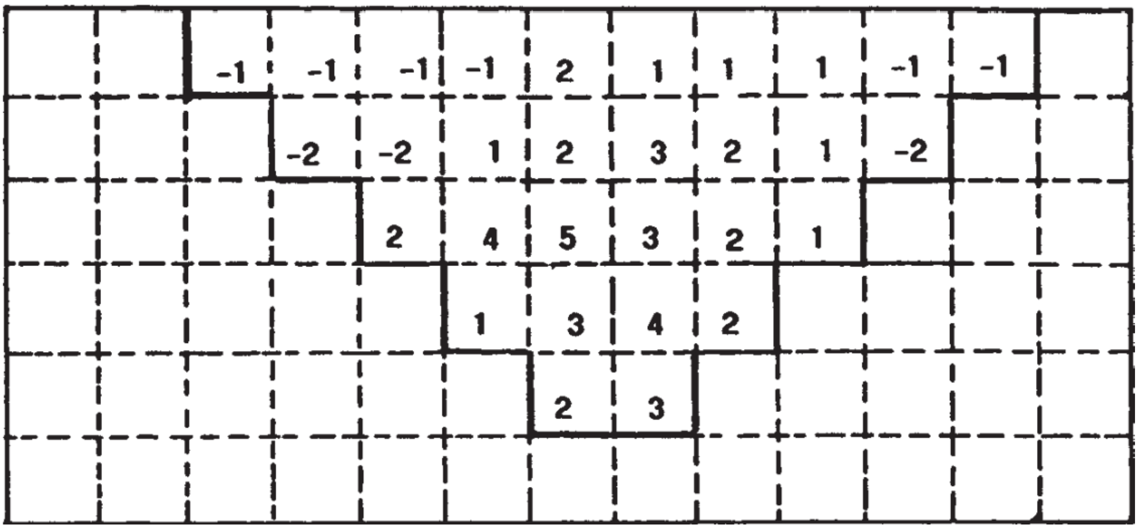
$$NPV = \frac{0.5}{(1.1)^1} + \frac{0.5}{(1.1)^2} + \frac{0.5}{(1.1)^3} + \frac{0.5}{(1.1)^4} + \frac{8}{(1.1)^5} = Rs. 6.55$$

(4) 2.25W1.75O→2.25W1.75O →2.25W1.75O →2.25W1.75O →1W3O

$$NPV = \frac{1.25}{(1.1)^1} + \frac{1.25}{(1.1)^2} + \frac{1.25}{(1.1)^3} + \frac{1.25}{(1.1)^4} + \frac{5}{(1.1)^5} = Rs. 7.06$$

7. An economic block model is shown below.

(a) Show the final pit layout using Floating Cone Technique.



(b) Determine the sequence in which the blocks should be mined as per the algorithm developed by Roman (1974).

-1	-1	-1	-1	-1	-1	2	1	1	1	-1	-1	-1
-1	-1	-2	-2	-2	1	2	3	2	1	-2	-1	-1
-2	-1	-1	-1	2	4	5	3	2	1	-2	-2	-2
-4	-3	-2	-2	-2	1	3	4	2	-1	-1	-2	-6
-6	-6	-5	-5	-4	-5	2	3	-2	-2	-3	-4	-5
-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6

(2 + 2 = 4 marks)

		-1 **(26)**	-1 **(17)**	-1 **(10)**	-1 **(7)**	2 **(1)**	1 **(2)**	1 **(3)**	1 **(5)**	-1 **(13)**	-1 **(21)**	
			-2 **(29)**	-2 **(18)**	1 **(11)**	2 **(8)**	3 **(4)**	2 **(6)**	1 **(14)**	-2 **(22)**		
				2 **(28)**	4 **(19)**	5 **(12)**	3 **(9)**	2 **(15)**	1 **(33)**			
					1 **(29)**	3 **(20)**	4 **(16)**	2 **(24)**				
						2 **(30)**	3 **(25)**					

8. Using the table shown below, determine

(a) the accounting rate of return for the initial capital invested and life of the project is 4 years.

Net income = -100,000 + 28,000 + 30,000 + 35,000 + 32,000 = 25,000

Average net return per year = $\frac{25,000}{4} = 6,250$

accounting rate of return = $\frac{6,250}{100,000} * 100 = 6.25\%$

(b) dynamic payback period using linear interpolation for discount rate of 8% per annum.

Time	0	1	2	3	4
Net Cash Flows	-100,000	28,000	30,000	35,000	32,000

(2 + 2 = 4 marks)

Time	Net cash flows	Present value	Left to pay
0	-100,000	-100,000	-100,000
1	28,000	25,925.93	-74,074.07
2	30,000	25,720.17	-48,354.90
3	35,000	27,784.13	-20,569.77

4	32,000	23,520.96	+2,951.19

Between 3-4 year, it will be paid.

dynamic payback period using linear = 3 + $\frac{20,569.77}{23,520.96} = 3.87\ years$

9. A mill concentrates, having 25% copper, is proposed to be sold at Rs. 1,25,000 per tonne. The grade of the deposit is 0.8% Cu and the overall cost of mining and milling is Rs. 2,520 per tonne of ore. At a recovery of 75%, estimate profit (in Rs./tonne) of concentrate.

(3 marks)

Grade of copper = 0.8% and Recovery = 75%

So, the copper content per tonne of ore = $\frac{0.8}{100} *$* 1000 = 8 kg

Now, copper recovered by mill = 8* $\frac{75}{100}$ = 6 kg

Here it is given that copper is 25 % in mill concentrate

It means that,

100 kg of concentrate have 25 kg copper

Then 1 kg of copper can be produced from $\frac{100}{25}$ = 4 kg of concentrate per tonne of ore

So, 6 kg of copper can be produced from 6 * 4 = 24 kg (0.024 tonne) of concentrate per tonne of ore

So, 1 tonne of concentrate produced from $\frac{1}{0.024}$ tonne of ore

So, Estimated profit (Rs/tonne) = 125000 - $\frac{1}{0.024} * 2520$ = 125000 – 105000 = 20000

Section B: Descriptive Answer Type Question (Answer in 2-3 lines/pointwise) – 15 marks

Answer all the questions and marks are assigned against each question

10. Name the stoping method/methods with following features:

a. Costliest: Square set stoping and Cut and Fill stoping

b. Cheapest: Shrinkage stoping or Sublevel stoping: Requires minimal support and backfilling.

c. Highest dilution: Sublevel caving and Block Caving

d. Least recovery: Room and pillar stoping; Sublevel stoping; Longwall

e. Highest productivity: Longwall mining

f. Lowest productivity: Square-set stoping

g. Early production but highest development: Sublevel caving and Room and Pillar, Sublevel Stoping

h. Subsidence unavoidable: Longwall or Block caving

i. Ore tie-up during stoping: Shrinkage stoping

j. Least exposure to unsafe conditions: Sublevel stoping

(5 marks)

11. Differentiate

(a) Salting and Sampling

Salting: A fraudulent practice of adding valuable material (e.g., gold, diamonds) to ore samples to mislead investors about the quality of a deposit.

Sampling: The process of collecting representative material from a deposit to assess its grade, composition, and value.

(b) Overhand and Underhand Stoping

Overhand Stoping: Mining progresses upward from a lower level. It is commonly used in deposits with good hanging wall stability.

Underhand Stoping: Mining progresses downward from an upper level. Used when the hanging wall is weak or unstable.

(c) Elastic Demand and Inelastic Demand

Elastic Demand: Demand changes significantly with a change in price. For example, luxury goods often have elastic demand.

Inelastic Demand: Demand remains relatively unchanged despite price changes. Essential goods like fuel or utilities often exhibit inelastic demand.

(d) Escrow Amount of Coal/Lignite and Mineral

Escrow Amount: A financial reserve or deposit held by a third party to ensure compliance with regulations (e.g., environmental or rehabilitation obligations) during mining operations.
For coal:
15 lakhs/hectare for OCP
2 lakhs/hectare for UGP
For mineral:
3 lakhs/hectare for mines where >175 employed in OCP or >75 employed in UG (such mines called category mine A)
2 lakhs/hectare for other than above category mine A (called B)

(e) Major Minerals and Minor Minerals

Major Minerals: Include coal, iron ore, bauxite, manganese, and other economically significant minerals regulated by central authorities.
Minor Minerals: Include sand, gravel, clay, and other locally significant materials regulated by state authorities.

(f) Project Year, Production Year, and Calendar Year

Project Year:
For both production and pre-production cash flow.
The first calendar year in which major expenditures occur is 1984. For the cash flow calculation this is selected as Project Year 1. Discounting will be done back to the beginning of year 1984.
The year marking the initiation or major milestones of a mining project.
The first year of production is 1991 which is eights year of the project. I doing the discounted cash flows and NPV calculations, they will be brought back to the beginning of the project.
e.g. 1, 2, 3,……
Calendar Year: The conventional year from January 1 to December 31 used for official and legal purposes.
For both production and pre-production cash flow. e.g. 1984, 1985, 1986,……
Production Year: The year when a project begins commercial extraction of minerals.

(6 marks)

12. Present a layout of cut and fill stope. Write down suitable conditions for its application and variants. What are the advantages and limitations. Mention the range of dimension used to space the levels.

(2 marks)

Layout of Cut-and-Fill Stope:
A typical cut-and-fill stope consists of:
A series of horizontal slices excavated sequentially.
After each slice, the void is backfilled with waste material or tailings to provide ground support.
Access drifts are provided for ore extraction, backfilling, and ventilation.

Suitable Conditions for Application:
Ore strength: moderate to strong.
Rock strength: weak.
Deposit shape: any, regular to irregular.
Deposit dip: usually steep but can be applied for flat dips also then it will be similar to longwall mining.
Size and thickness: fairly large extent, thin to thick (2–30 m).
Ore grade: high but uniformity can be variable.
Depth: practiced up to 2.5 km.
Orebody Characteristics: Steeply dipping and irregular deposits.
Moderate to high-grade ore with limited dilution tolerance.
Rock Mass Stability: Poor ground conditions requiring continuous support.
Environmental Constraints: Where minimal subsidence is critical.

Variants of Cut-and-Fill Stoping:
Cut and fill with flat back – (i) Conventional; (ii) Mechanized
Cut and fill with inclined slicing

Longwall cut and fill stoping.
Post & pillar – cut and fill stoping
Stope-drive cut and fill stoping:
Overhand Cut-and-Fill: Mining progresses upward; suitable for weak ore and hanging wall conditions.
Underhand Cut-and-Fill: Mining progresses downward; used when the hanging wall is unstable.
Post-pillar Cut-and-Fill: Leaves ore pillars for added support.

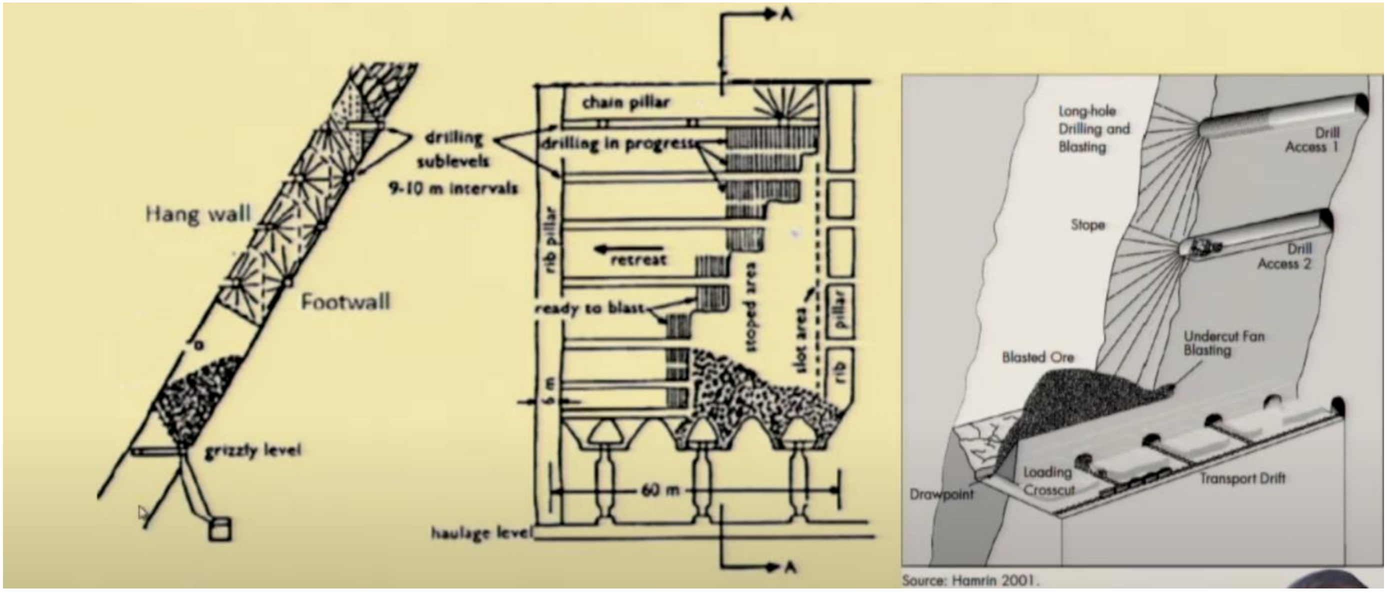
Advantages:
Mining cost – relative cost 60%.
Cost of backfilling – up to 50% of the total mining cost.
Operational skill – requires skilled labor. It is more labor intensive. ● Working atmosphere: at depth wet filling may create humidity problems.
High recovery and minimal dilution.
Flexible for irregular ore bodies.
Provides continuous support with backfill, enhancing safety.
Suitable for weak or fractured ground.

Limitations:
High cost due to backfilling and development.
Slow production rate compared to mechanized methods.
Requires careful monitoring of ground conditions.

Dimension Range for Level Spacing:
Typically 10 to 30 meters, depending on deposit thickness, rock mass quality, and equipment size.

13. Present a layout of sublevel stoping. Write down suitable conditions for its application and variants. What are the advantages and limitations. Mention the range of dimension used to space the levels.

(2 marks)



Layout of Sublevel Stoping:
A typical sublevel stoping layout includes:
Orebody: A steeply dipping deposit divided into stopes.
Sublevels: Horizontal levels spaced at regular intervals for drilling and blasting.
Access: A main haulage drift at the bottom for ore collection and vertical raises or ramps for access.
Extraction: Blasted ore falls to the base of the stope and is collected through drawpoints or ore passes.

Suitable Conditions for Application:
Orebody Geometry:

Steeply dipping (>50°) and tabular or lenticular deposits.
Regular shape and continuous orebody.
Rock Mass Strength:
Competent ore and host rock to maintain stope walls without collapse.
Ore Grade:
Medium to high-grade ore, requiring cost-efficient extraction.

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sublevel stoping can be classified as under:
● Sublevel stoping with benching – with the use of conventional drills.
● Blasthole stoping – with the use of blasthole drills.
● Big/large blasthole stoping – with the use of large diameter drills such as DTH drills.

Variants of Sublevel Stoping:
Open Stoping: Classic variant with no backfill.
Sublevel Open Stoping: Includes sublevels for drilling and blasting optimization.
Post-pillar Sublevel Stoping: Leaves pillars between stopes for added ground support.

Advantages:
High production rate with mechanized drilling and blasting.
Cost-effective for large, continuous deposits.
Minimal dilution and high recovery.
Relatively safe due to remote drilling and blasting.
Limitations:
Limited to strong rock conditions; weak rocks require additional support.
Inefficient for irregular ore bodies.
Initial development is capital-intensive.

Dimension Range for Level Spacing:
Typically 20 to 50 meters, depending on orebody thickness, equipment, and drilling reach.

Section C: Descriptive Answer Type Question (Answer in 2-3 lines/pointwise) – 21 marks

Answer all the questions and each question carries 1 mark only

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14. Mention the basic line items in a pre-production cash flow table.

1. Project year 2. Calendar year 3. Capital expenditures: Total 4. Property acquisition 5. Royalties 6. Exploration 7. Development 8. Mine/mill buildings 9. Mine/mill equipment 10. Property tax 11. Working capital 12. Total capital expenditures 13. Cash generated due to tax savings 14. Exploration 15. Development 16. Depreciation 17. Property tax 18. Total cash generated 19. Net cash flow

15. Mention the basic line items in a production period cash flow table.

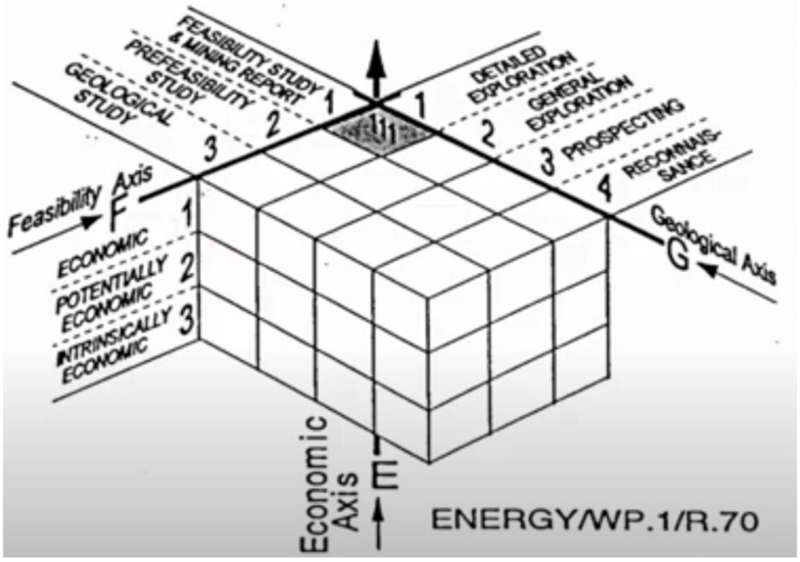
1. Production year 2. Project year 3. Calendar year 4. Revenue 5. Royalty 6. Net revenue 7. Mining cost 8. Processing cost 9. General cost 10. Property tax 11. Severance tax 12. Depreciation 13. State income tax 14. Net income after costs 15. Depletion 16. Taxable income 17. Federal income tax 18. Profit 19. Depreciation 20. Depletion 21. Cash flow 22. Capital expenditures 23. Working capital 24. Net cash flow

16. Taylor has provided some useful insights into deciding the appropriate mine life. Summarize the factors which should be considered. Develop the reasoning as to why one might expect the life of a mine to be related to the cubic root of the ore tonnage.

Factors to Consider for Mine Life (Taylor's Insights):
Ore Tonnage: Larger ore bodies typically have longer mine lives.
Grade and Value: High-grade ore may favor shorter mine life due to faster recovery.
Market Demand: Prices and demand influence production rates and mine life.
Mining Costs: High operational costs may extend mine life for economic viability.
Capital Investment: Significant investment often justifies extended mine operation.

Reasoning for Relation to Cubic Root of Ore Tonnage:
The rate of ore extraction often scales with the mine area or volume, which depends on the mine's physical dimensions (length, width, depth).
Since volume is proportional to the cube of a linear dimension, mine life (time for full extraction) approximates the cubic root of ore tonnage when production rates are proportional to mine size.

17. Discuss UNFC reserve classification system.



18. Discuss the objectives and key provisions of National Mineral Policy 2019.

Objective
Effective regulation
Transparency
This could significantly address the issues of project affected persons especially those residing in tribal areas.

Key provisions
Industry status
Right to first refusal
• private sector
Transportation
Fund District Mineral Fund
• export import
• inter-generational equity
Regulation

19. Mention the purpose of mine valuation.

Acquisition
Taxation
Financing
Regulatory Requirements

20. Discuss the methods of investment appraisal.

Static Methods
Cost Comparison Method
Profit Comparison Method
Average Rate of Return Method
Static Payback Period Method
Discounted Cash Flow Methods
Net Present Value Method Annuity Method

Internal Rate of Return Method
Dynamic Payback Period Method

21. Mention different kinds of mineral price quotation and factors affecting mineral price.

kinds of price quotation υ From the point of production up to that of sale, each stage involves addition of some cost. As a result, price of a mineral commodity depends on the stage at which it is sold. Broadly, the price includes one or more of the different costs incurred , i.e. cost of production, cost of transportation, cost of loading, etc. while cost of loading depends on the type of transports, the cost of transportation depends on the distance up to which the commodity is transported for sale. Accordingly, prices are usually quoted by traders with reference to the type of transport and also with reference to the market place.

Different kinds of price quotations are:
P.M.V. price: Pits mouth value (cost of mining, dead rent, royalty, other taxes for mining)
Ex-Pit of Ex-mine price (ore brought to the pits mouth is broken to salable size and prepared for sale)
F.A.S. price: free alongside ship (mineral transported to the port side, this price includes the ex pit price, the cost of transportation, cost of unloading, maintenance of dump yard, cost of export duties etc.)
F.I.D.price: free into container depot (transported to railway siding but to container depot)
F.O.R. price: free on rail (loaded into railway wagons)
F.O.B price: free on board (PMV, transportation to port, and loading into the vessel)
F.O.B.T price: free on board and trimmed
C. I. M price: cost, insurance and freight (FOBT price are added in the insurance and ocean freight charges
At the micro level
Levels of capacity utilization
The sales volumes at any given time
The size of the order at any given consumer
The extent of long standing relations with a consumer
The potential that is seen to exist in the given market
The nature of the product
At macro level
Reserve and cut off grade
Protection to consumers
Stocks/Demands
Geological mode of occurrence
Inflation and input cost
Performance of industry
Unions and associations (MMTC, CSO, OPEC)
Statutory price At macro level υ Protection to domestic producers
International price
Sales promotion
Obligations or IMF
Exchange rate
Taxation policy
Market forces
Interrelation amongst commodities
Mineral price index

22. Discuss the steps taken by Indian government to ease the process of licensing, clearances and approvals required for operating a mine.

The Indian government has streamlined mining approvals by implementing a **single-window clearance system SWAYAM Portal** for faster processing, digitizing the auction process for transparency, and introducing **online environmental and forest clearances**. Policies like the MMDR Amendment Act have simplified lease renewals, while initiatives like **PARIVESH** have integrated environmental clearances.

23. Discuss methods of mineral conservation for sustainable development.

Methods of conservation
Continuous search for new sources of raw materials
Choice of correct mining method υ Utilization of unmarketable, by ore dressing
Use of re-use of scrap
Recovery of all associated elements
Economy in the use of mineral
Finding new substitutes
Newer methods of mining and utilize thinner ore
Prohibiting non-essential use
New ideas for conservation and promotion
Stowing
Coal washeries
Efficient production/ clean coal production
Recovery of by products
Blending

24. Discuss coal mine closure plan.

A **coal mine closure plan** in India focuses on sustainable closure of mines to minimize environmental and social impacts. It comprises two components:
Progressive Mine Closure Plan: Continuous reclamation and rehabilitation during the mining phase, including afforestation and restoring disturbed land.
Final Mine Closure Plan: Activities after mining ends, such as sealing shafts, removing infrastructure, and ensuring ecological stability.
Operators must deposit funds in an **escrow account** to finance closure activities. The plan also includes socioeconomic measures, like reskilling affected workers and community development initiatives, to mitigate post-mining impacts.

25. Briefly describe the various stope selection techniques and their applicability.

Typical considerations to be weighted in selecting mining methods:
● Maximize safety
● Minimize cost (bulk mining methods have lower operating cost than selective)
● Minimize the schedule required to achieve full production
● Optimize recovery (80% or more of the geological reserves)
● Minimize dilution (20% or less of waste rock that may or may not contain economic minerals)
● Minimize stope turn around (cycle time for various unit operations)
● Maximize mechanization
● Maximize automation (employment of remote controlled equipment)
● Minimize pre-production development (top down versus bottom up mining)
● Minimize stope development
● Maximize gravity assistance (underhand versus overhand)
● Maximize natural supports
● Minimize retention period (open stoping versus shrinkage)
● Maximize flexibility and adaptability:
– Based on size, shape, and distribution of target mining areas
– Based on distribution and variability of ore grade size, shape, and distribution of target mining areas
– To sustain the mining rate for the mine life
– Based on access requirement
– Based on opening stability, ground support requirements, hydrology (ground water and surface runoffs), and surface subsidence.

Stope Selection Techniques:

Geological Factors:

Based on orebody geometry, grade distribution, and rock stability.
Applicable for irregular deposits requiring selective methods like cut-and-fill stoping.

Economic Factors:
Evaluates costs, ore recovery, and dilution.
Sublevel caving suits large, low-grade deposits for cost-effective mining.
Environmental Considerations:
Considers subsidence and surface impacts.
Longwall mining applies where planned subsidence is acceptable.
Technical Factors:
Depends on equipment, safety, and access.
Shrinkage stoping works well for steep, competent orebodies.
Production Requirements:
Determines productivity and ore tie-up.
Block caving is ideal for high-tonnage, mechanized mining.
The technique chosen depends on deposit characteristics, operational constraints, and project goals.

26. What are the assumptions that need to be made in the mine life-plant size determination?

– Assumption 1. Cost-recovery values.
– Assumption 2. Commodity price.
– Assumption 3. Destination options.
– Assumption 4. Mill cutoff grade.
– Assumption 5. Yearly product mix.

27. Group **(a)** orebodies, country rocks based on their stability and **(b)** orebodies based on their thickness and dip

With regard to stability of ores and country rocks, no index is available which determines the permissible value of the exposure – time and span.
However, this may be divided into five groups as outlined below
● Very unstable ores and rocks: The rocks which do not allow any exposure of roof and walls and need advanced timbering; such rocks and ground are: quicksand, loose and water saturated strata etc.
● Unstable ores and rocks: In this type of strata the roof/back needs a strong support immediately after its exposure.
● Medium stability ores and rocks: This type of strata permit exposure of the roof over a comparatively large area and requires support if the roof exposure is to be allowed for a considerable time period.
● Stable ore and rocks: In these strata roof and side exposures can be allowed for a considerable time without support. However, some patches may require some support.
● Very stable ores and rocks: This type of strata can allow exposure for sufficient time and space without causing caving. Strata of this kind are rarely encountered comparing the same with the previous two groups.

To determine stability of the strata, it is important to know the nature of caving i.e.
● Caving occurring at once over a large area
● Caving gradually in small sections by inrush of separate lumps or layers
● Caving after giving certain indications, or it occurs without warning/unexpectedly.

There is no standard classification with regard to thickness of ore, however, a thickness grouping could be as under:
Very thin deposits: thickness less than 0.7 m.
Thin deposits: thickness range 0.7–2 m.
Medium thick deposit: thickness range 2–5 m.
Thick deposit: thickness range 5–20 m.
Very thick deposit: thickness exceeding 20 m.
Usually the following classification holds good, with regard to dip of a deposit:
Flat dipping: 0° to below 20°
Inclined dipping: from 20° to below 50°
Steeply inclined dipping: exceeding 50°

28. List factors governing choice of a mining method.

Factors governing choice of a mining method
Shape and size of the deposit
Thickness of deposit
Dip of the deposit
Physical and mechanical characteristics of the ore and the enclosing rocks
Roof Pressure:
Presence of geological disturbances and influence of the direction of cleats or partings
Degree of mechanization and output required
Ore grade and its distribution, and value of the product
Depth of the deposit
Presence of water
Presence of gases
Ore & country rock susceptibility to caking and oxidation

29. List the desirable features a stoping method should be able to render, if selected.

1. The facility it gives for rock fragmentation: This may be examined by:
● Ease it can impart during drilling operation
● Effectiveness of the blast
● Facility it can provide for the crew, machines and tools to be used for rock fragmentation purposes
● Drilling is easy when workers get a sufficient and firm foothold, and there is no obstacle for installation, operation and shifting of the equipment used. Blasting becomes more effective, if use of Earth's gravity can be made while sequencing the blast.
2. Output: Concentrating over few working faces with the application of the resources in terms of personnel, machines, equipment, services and supervision can yield better productivity i.e. output/person/shift. Also selecting of proper matching equipment for carrying out various unit operations is essential for better results.
3. Rate of progress: This aspect is important not only from the viewpoint of output but for the proper stability of ground. A fast rate of progress can reduce expenses on supports and mine services. Many methods have been designed for faster rate of development, stoping and production.
4. Maximum recovery: A good mining system should aim for maximum ore recovery in terms of tonnage and grade. Dilution should be minimum. The bulk mining methods may yield high output but often they are associated with higher dilution (sec. 16.4.2–3). Requirement of large number of pillars also results in poor overall recovery.
5. Clean mining: This means not contaminating the ore either with the filling waste or with the in-situ waste rock from the roof and sides of the stopes. Clean mining is possible if the method selected allows selective mining. Faster rate of progress can also yield a clean mining as in this case roof exposure will be minimum, and thereby, ore contamination by the falling fragments from the roof will also be very little.
6. Easy access within stope: This differs from one stoping method to another; however, mechanized mining can allow easy movement of personnel, machines, equipment and mine services.
7. Energy & material consumption: The methods which involve higher consumption of timber or other materials for the purpose of supporting the stope workings, are costlier. Another consumable item is energy/unit of output, which is required to carryout different unit operations and services such as ventilation, pumping, illumination etc.
8. Ore handling: Quick handling/removal of the broken muck from the stopes is essential for safety and productivity, and therefore, methods allowing this facility should be preferred.
9. Proper filling/packing: If mining method warrants back filling of the worked out space, method's adaptability to easy and effective filling should be looked into. A minimum void in the worked out space after filling should be aimed at.
10. Ore winning postures and danger from roof fall/caving and muck rolling: The ore can be extracted either by underhand, breasting or overhand stoping. Mining and transport are difficult in underhand stoping and easier in overhand stoping as shown in figures 16.1(a) & (b). In underhand stoping working

below an exposed roof involves danger of roof fall or caving. To minimize this danger, roof bolting frequently becomes essential.

30. Classify underground stoping methods.

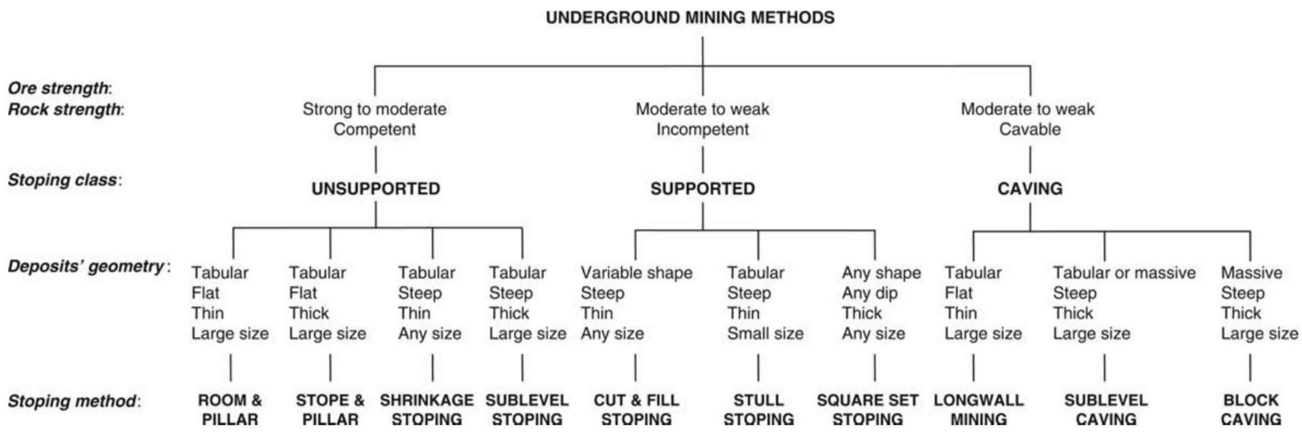


Figure 16.2 Classification of stoping methods – Based on deposit characteristics.

31. Discuss model and design parameters considered while designing stope.

Model parameters

This includes the parameters such as orebody profile in plan (at various horizons) and sections (along various latitudes and departures) which, in turn, depicts the orebody's thickness and dip.

It means the stope is to be designed for the orebody presented in this manner

Design parameters

In order to mine out a stope, the following provisions need to be made:

a. Access to the stope for workers, machines, equipment and mine services (Service Raise or Manpass).

b. Creation of free face for the stoping operations to start with (Slot).

c. Rock fragmentation by way of drilling and blasting the blast holes (Drill drives/ sublevels or crosscuts).

d. Handling fragmented ore muck (Extraction level layout).

e. Limiting and isolation of stope size or dimensions (Pillars of various kinds).

f. Approaching geological features.

32. Discuss the different pillar types during different stoping methods.

Pillar Types in Different Stoping Methods:

Room-and-Pillar Stoping:

Yield Pillars: Deform under stress to absorb load and maintain stability.

Barrier Pillars: Large pillars left to protect adjacent mining areas or infrastructure.

Chain Pillars: Smaller pillars connecting rooms, maintaining localized support.

Sublevel Stoping:

Crown Pillars: Left to support the overlying rock between the stope and surface or adjacent workings.

Rib Pillars: Vertical pillars left between stopes for structural stability.

Shrinkage Stoping:

Hanging Wall Pillars: Left to prevent collapse of weak hanging walls.

Footwall Pillars: Used to ensure stability in weak footwalls.

Cut-and-Fill Stoping:

Temporary Pillars: Often created with backfill to support sections of the stope during mining.

Rock Pillars: Natural ore or rock sections left unmined to provide immediate support.

Longwall Mining:

Gate Road Pillars: Support the main entries and maintain stability during extraction.

Barrier Pillars: Protect critical infrastructure or unused portions of the seam.

Caving Methods (e.g., Block Caving):

Protective Pillars: Left around active cave zones to prevent unintended collapses.

Pillar design varies by method and depends on rock mass strength, stress conditions, and operational requirements.

33. Discuss ground control issues associated with different stoping methods.

Ground Control Issues in Different Stoping Methods:
Cut-and-Fill Stoping:
Stability of hanging walls and backfill is critical.
Requires careful monitoring and support to avoid wall failures.
Shrinkage Stoping:
Risks include rock bursts and overbreak in steep orebodies.
Hanging wall and footwall stability is vital as ore is temporarily stored in the stope.
Sublevel Stoping:
Challenges include stress redistribution causing rock failures.
Proper pillar design and stope sequencing mitigate ground pressure issues.
Room-and-Pillar Stoping:
Pillar collapse can lead to subsidence or loss of access.
Requires optimized pillar dimensions for stability and recovery.
Block Caving:
Induced subsidence and surface cracks are common.
Effective fragmentation and cave management are needed to control ground movement.
Longwall Mining:
High subsidence potential due to complete seam extraction.
Requires robust support systems to handle dynamic ground stresses.
Ground control depends on site-specific conditions like rock mass quality, stress regime, and mining geometry.

34. Discuss liquidation and their techniques.

MINE LIQUIDATION
In underground mines the worked out space is the space created as a result of exploitation of a deposit. Every such space and its size have their effects on the surrounding rock mass. Liquidation is the systematic abandoning by way of achieving the desired changes in the state and the extent of the influencing zone corresponding to the size of the worked out spaces. It is undertaken not only to ensure ultimate safe and economic exploitation of the deposit but also to keep in mind the quantitative and qualitative aspects of conservation of a mineral wealth.
Liquidation of the stopes of different types
The size of worked out space and the period for which it can keep standing depends upon the strength characteristics (dynamic as well as static) of the orebody and its surrounding rock mass. Sometimes difficulties that arise while working the lower levels can also be attributed as a function of these characteristics. Liquidation operation can be grouped as per the stoping methods outline below:

- Liquidation of caving stopes (fig. 16.41(i)(c))
- Liquidation of supported/filled stopes (fig. 16.41(i)(b))
- Liquidation of open stopes (fig. 16.41(i)(a))
- Liquidation of standing stopes.

Best Wishes!
Question Paper Ends Here